SMART HACKS 2026

SWASTHYASAATHI AI

- **Problem Statement ID –** 25049
- **Problem Statement Title-** AI-driven Public Health Chatbot for Disease Awareness
- **Theme-** MedTech/BioTech/HealthTech
- **PS Category-** Software
- **Team Name-** ZenZeal 18

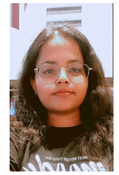
Mentor :- Prof.Tarun Dev

Assistant Proffesor, B.M.R.E. Department

Subject Expertise: Renewable Energy, 3D Printing (Additive Manufacturing), Robotics & Automation, Artificial Intelligence and Machine Learning, Bio-materials



➤ Team Details:-

Role	Name	Batch	Roll no.	Photo
Team Leader	Aditya Anand Kashyap	2k23	23BR02	
Team Member	Anuradha Kumari	2k23	23BR40	
Team Member	Akash Chandra Verma	2k23	23IT02	
Team Member	Raiyan Hasan	2k23	23IT57	

SWASTHYASAATHI AI

Problems

- **Delayed treatment** due to poor early symptom awareness.
- **Language barriers limit** understanding of health information.
- **Missed vaccinations** because of no reminders and tracking.
- **Health misinformation** spreads quickly through social media.
- **Late outbreak alerts** in local communities.
- **Limited doctor access** in **rural** and **semi-urban areas.**

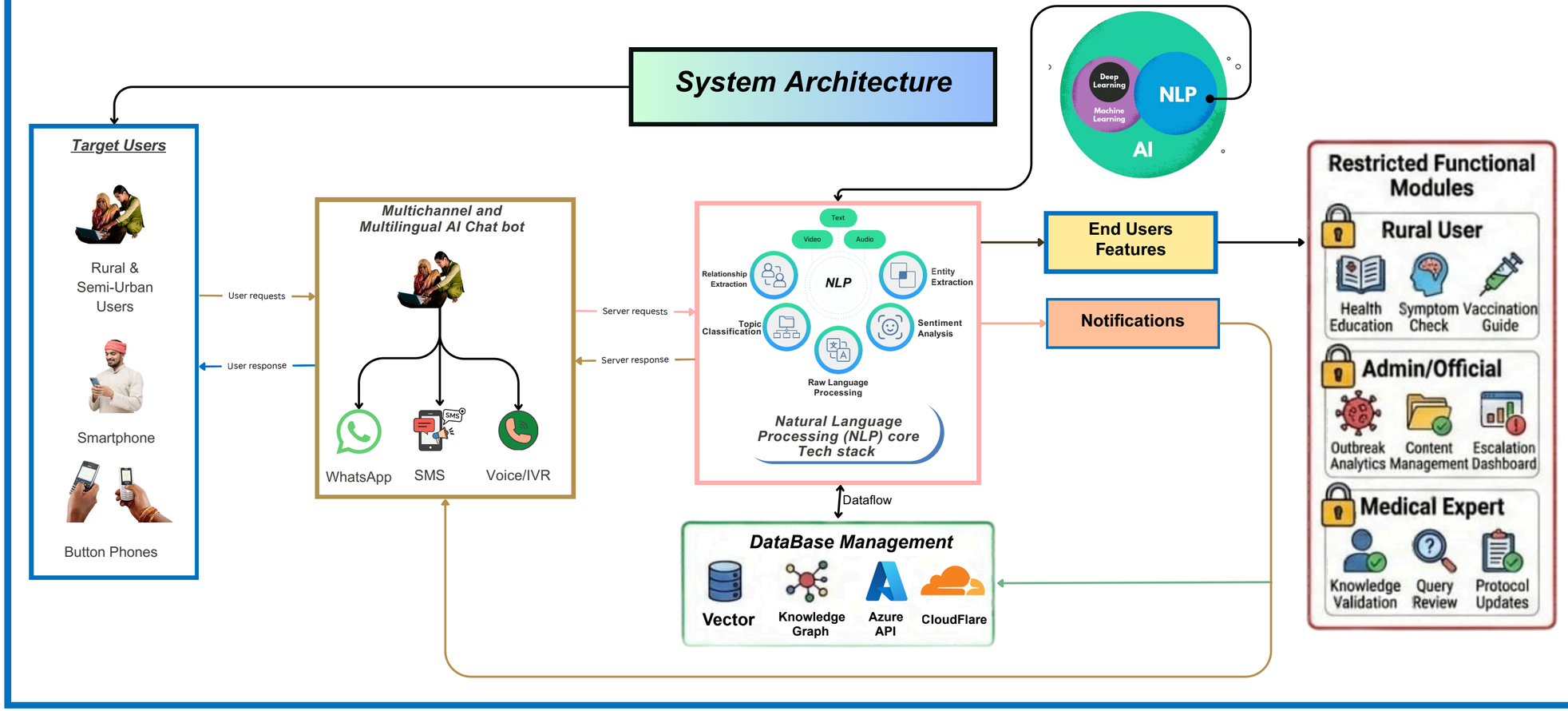
Proposed Solution

1. **Multilingual AI chatbot** for easy health guidance in local languages.
2. **Early symptom awareness support** with safe, preventive advice.
3. **Automated vaccination reminders** for children and mothers.
4. **Myth-busting health education** using **verified** government data.
5. **Real-time outbreak alerts** based on **district-level data.**
6. **24/7 accessible** via **WhatsApp & SMS** for rural reach.

Innovation & Uniqueness

- Voice-Enabled Chatbot
- Multilingual Support
- Works on SMS & WhatsApp
- Vaccination Reminders
- Outbreak Alerts
- Offline Knowledge Access

System Architecture



THE IMPACT: MEASURABLE OUTCOMES

- +45% AWARENESS REACH
- 90% VACCINATION COMPLIANCE
- -30% OUTBREAK CASE RATE

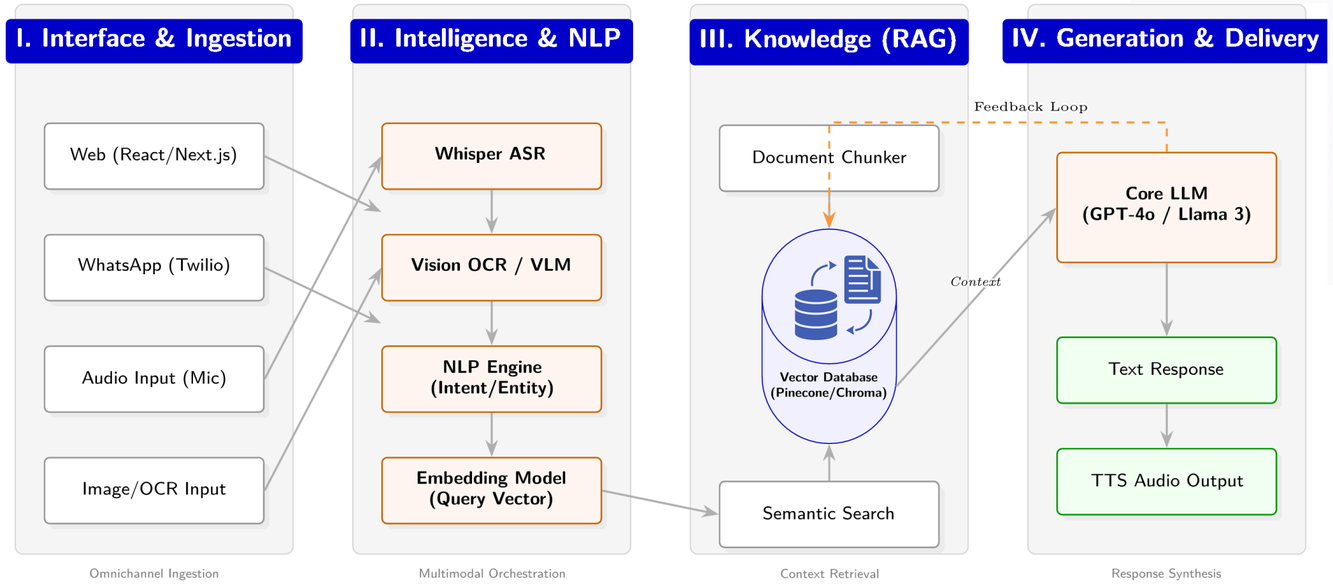
Improved health metrics & empowered rural communities

TECHNICAL APPROACH

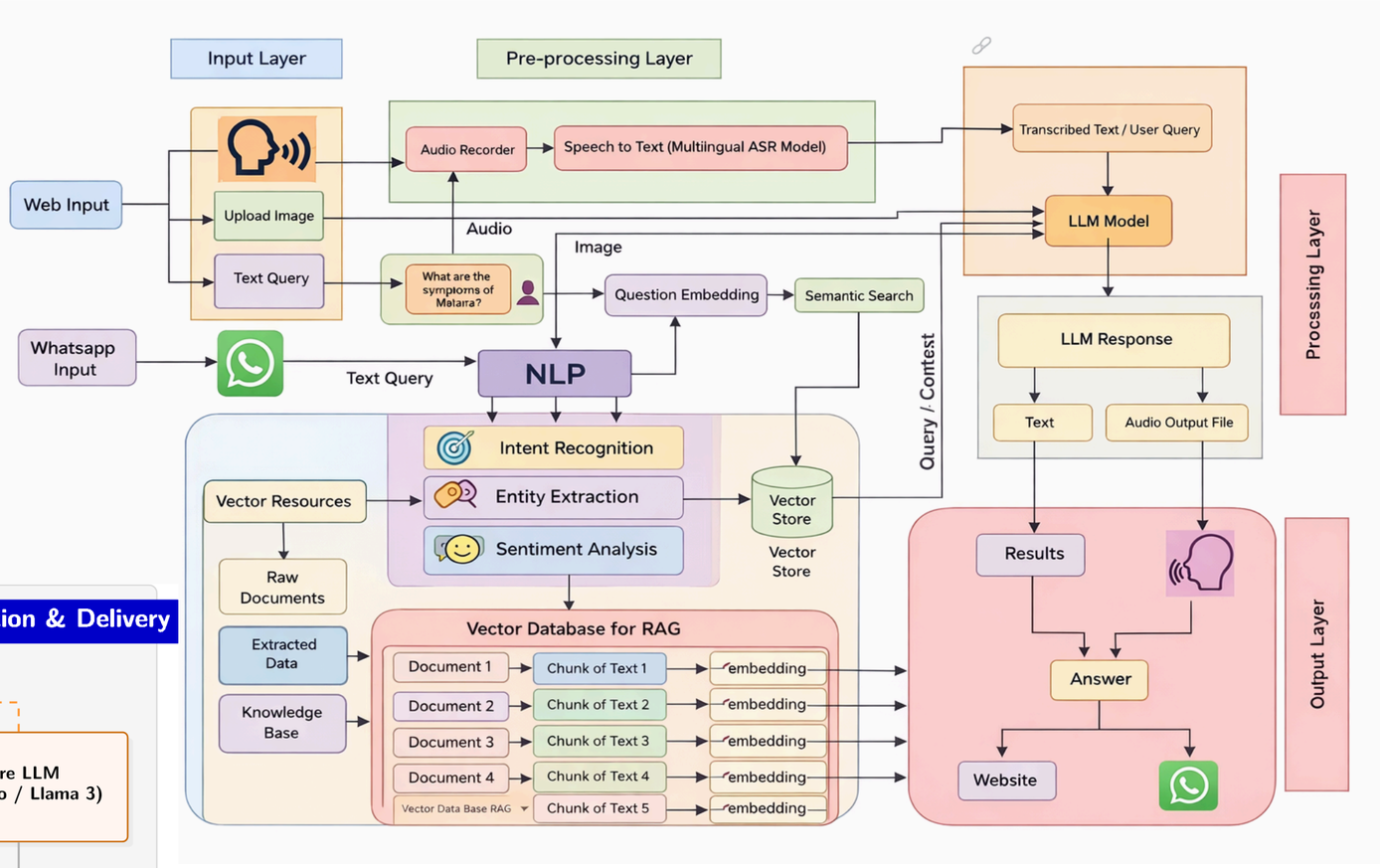
Technical Stack

- **Input Channels:** Web, WhatsApp,sms
- **Input Types:** Text Query, Upload Image, Audio
- **Pre-processing:** Audio Recorder, Multilingual ASR Model (Speech to Text)
- **NLP Engine:** Intent Recognition, Entity Extraction, Sentiment Analysis
- **Retrieval System:** Semantic Search, Question Embedding, Vector Store
- **Vector Database for RAG:** Raw Documents, Extracted Data, Knowledge Base, Document Chunking, Embeddings
- **Core Intelligence:** LLM Model
- **Output Modalities:** LLM Response (Text, Audio Output File)
- **Delivery Endpoints:** Website, WhatsApp

AI(NLP) BACKEND FLOW

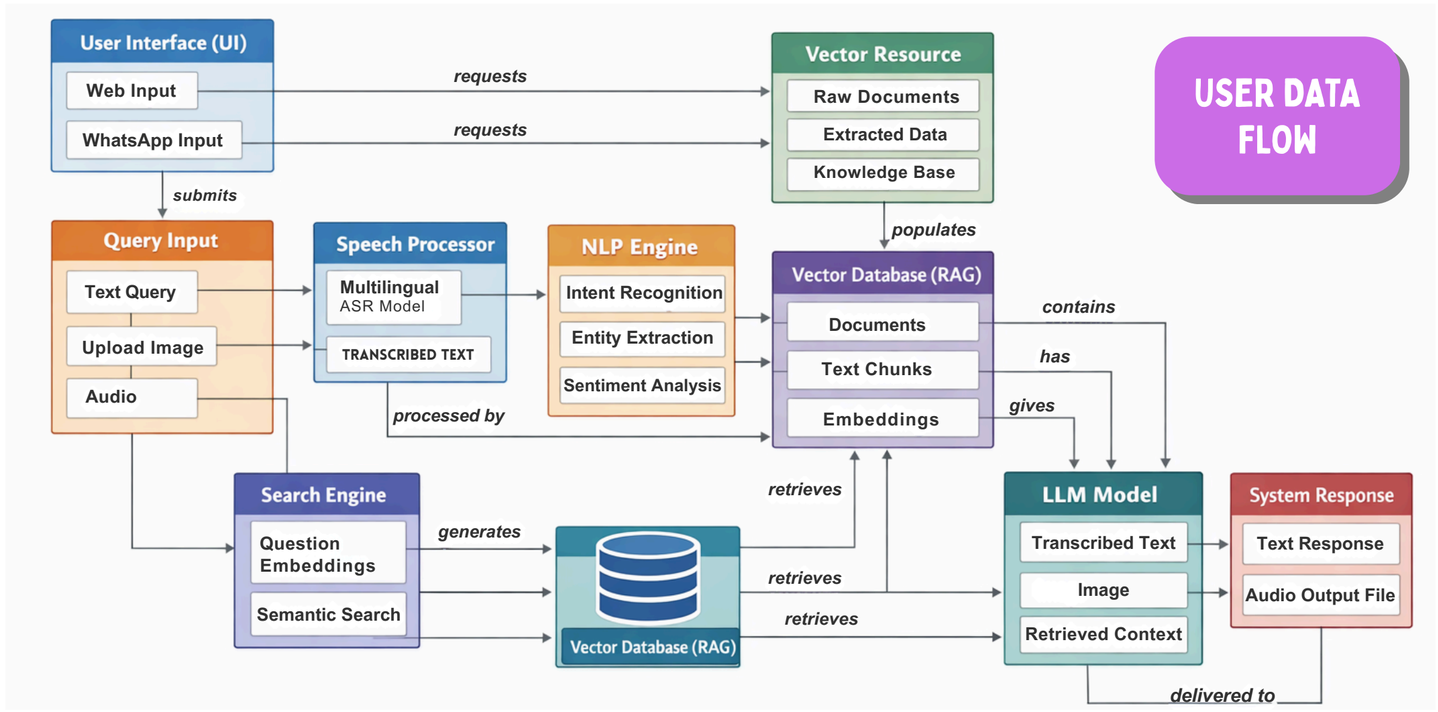


Technical Architecture





DATA FLOW AND USER EXPERIENCE

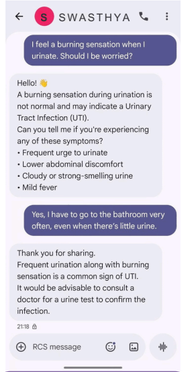


User Experience Across Platforms :-

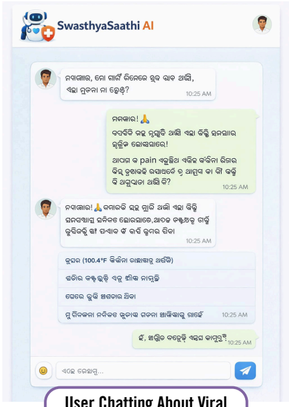




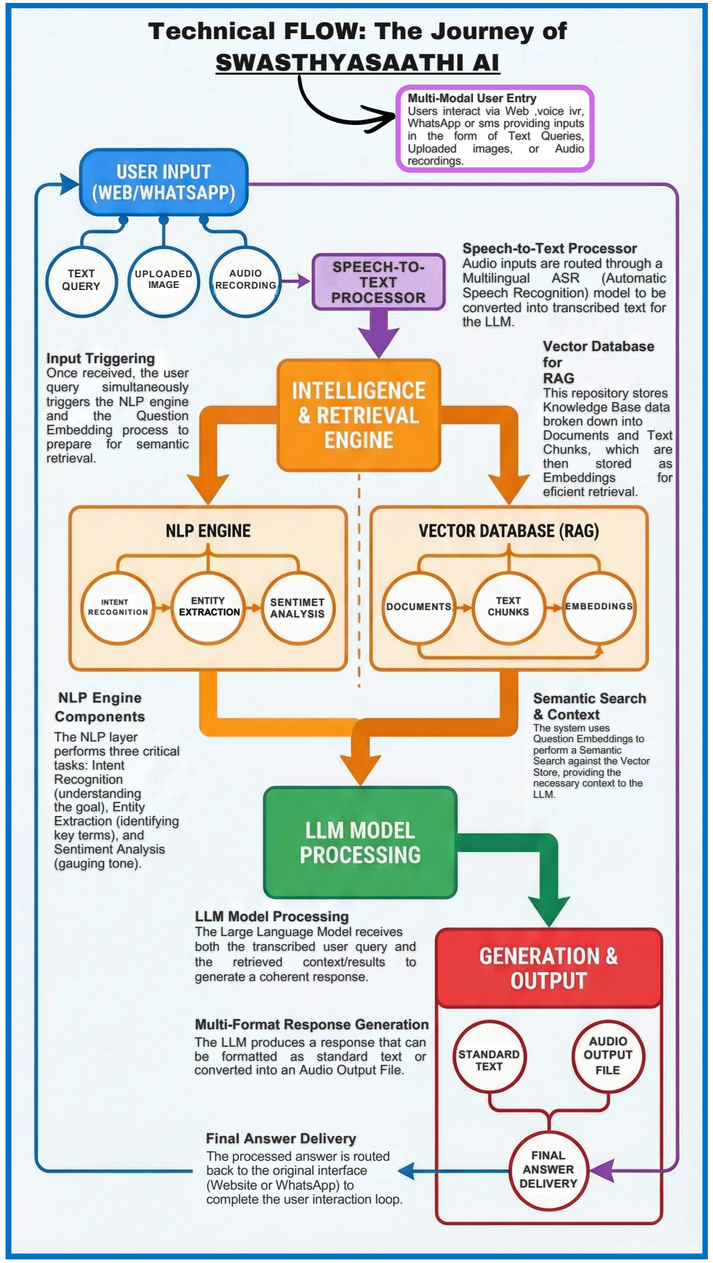
User Chatting About Dementia on WhatsApp in Hindi



User Chatting About UTI Disease Awareness via SMS



User Chatting About Viral Fever in Odia on the Web App



FEASIBILITY AND VIABILITY

Feasibility

- Feasible with existing AI/ML & Govt. APIs, ensuring cost-effective implementation.
- Scalable and accessible through cloud deployment & multilingual support.

Viability

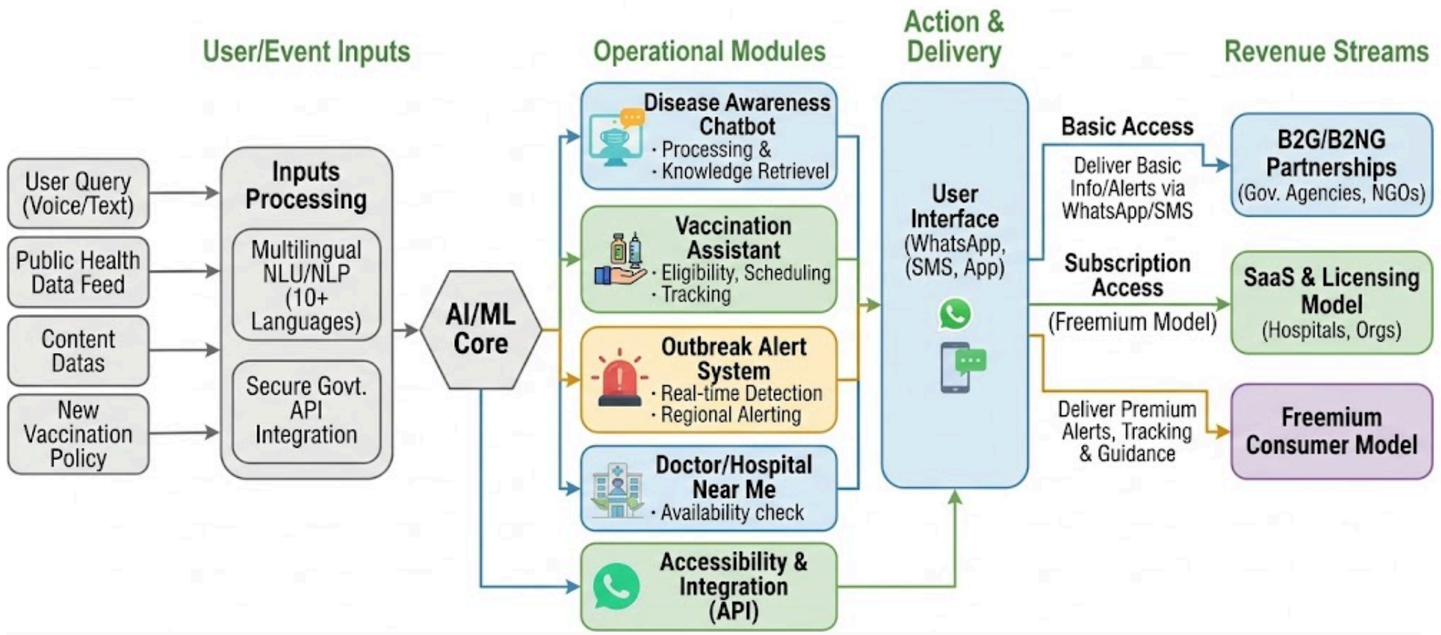
- High social impact by improving awareness, vaccination, and outbreak preparedness in rural & semi-urban areas.
- Supported by Govt/NGOs with a low-cost, scalable, and sustainable digital model.

Challenges

- Digital divide may limit rural reach.
- Requires accurate multilingual & voice support for low-literacy users, and strict data privacy to secure health info & build trust.

Business Potential

- **B2G & B2NG Partnerships** – Collaborate with Govt. health departments, NGOs, and public health programs for disease awareness, vaccination drives, and outbreak management.
- **SaaS & Licensing Model** – Provide API access and chatbot deployment to hospitals, clinics, and health organizations for subscription-based revenue.
- **Freemium Consumer Model** – Free access to basic health info via WhatsApp/SMS; premium features like vaccination tracking, outbreak alerts, and doctor suggestions generate monetization.



Key Differentiators:

- Multilingual Support (10+ Languages)
- Voice-First Inclusion
- Government Integration Ready
- Offline Capability
- Real-time Outbreak Monitoring

Target Impact:

- Bridge Healthcare Access Gap
- Improve Preventive Healthcare
- Support Rural Health Initiatives
- Enhance Public Health Response
- Drive Digital Health Adoption

Success Metrics:

- User Engagement & Retention
- Health Outcome Improvements
- Partner Adoption Rate
- Revenue Growth
- Market Penetration

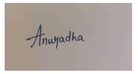
RESEARCH AND REFERENCES

- World Health Organization (WHO) – Global Health Data
- Ministry of Health & Family Welfare (MoHFW), India
- National Digital Health Mission (ABDM), Govt. of India
- Integrated Disease Surveillance Programme (IDSP)
- Grand View Research – Digital Health Market Report
- AIKosh – National Health Data Repository Initiative

➤ Key Supporting Statistics

- **India has a doctor-to-population ratio of ~1:834, with rural areas facing significant shortages (MoHFW / WHO data).**
- **India's digital health market is projected to grow at 25%+ CAGR, driven by AI and telemedicine adoption (Grand View Research).**
- **Over 65% of India's population lives in rural areas, where access to quality healthcare remains limited (Government of India Census data).**

"We agree to abide by all rules and policies set forth by MITMAAI. The decision of MITMAAI selection team will be binding and cannot be challenged. We also agree that, before filing any patent applications or forming any startup based substantially on their winning project, we will first consult with MITMAAI/the Mentor regarding strategy, protection, and commercialization, and will consider in good faith offering MITMAAI/the Mentor an appropriate advisory or equity role as may be mutually agreed in writing."

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Live Demo

SWASTHYASAATHI demo video

SWASTHYASAATHI Vercel app